

Disinformation analysis YouTube

Maggie Mazza, Casper Polhaar

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Setup

```
#load neccessary packages
library(tidyverse)
library(tinytex)
library(tidytext)
library(knitr)
library(corr)

# Read the datasets
# (Ensure these files are in your working directory)
video_meta <- read_csv("youtube_haitian_disinformation_video_meta.csv")
comment_meta <- read_csv("youtube_haitian_disinformation_comment_reply_metadata.csv")

# View the main dataset
glimpse(video_meta)
```

```
## Rows: 23
## Columns: 13
## $ ...1      <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ video_id  <chr> "OVS99PXbbw", "39voRIZciVI", "4IGd0BrrXoI", "f1RF50QrU~
## $ title     <chr> "Why Trump keeps pushing debunked claim about Haitian i~
## $ description <chr> "University of Virginia Center for Politics director La~
## $ date      <dtm> 2024-09-14 22:19:34, 2024-09-15 17:50:19, 2024-09-11 0~
## $ tags      <chr> "[ 'politics' 'us news' 'immigration' 'springfield ohio'~
## $ views     <dbl> 312222, 320834, 1858436, 43414, 236924, 1342405, 261990~
## $ likes     <dbl> 4327, 5938, 27856, 546, 5363, 35919, 4575, 5830, 8761, ~
## $ dislikes  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ comments  <chr> "[ 'What presidential candidate continues rhetoric that ~
## $ channel_title <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Fo~
## $ transcript <chr> "[ { 'duration': 1.335, 'start': 0.133, 'text': 'The mayo~
## $ transcript_str <chr> "The mayor of Springfield, Ohio, issued a public plea s~
```

Data preparation and comment counting

```
# Count comments per video
comment_counts <- comment_meta %>%
  group_by(video_id) %>%
```

```

summarise(comment_count = n(), .groups = 'drop')

# Merge, clean, and prep the dataset
video_prep <- video_meta %>%
  # Merge the comment counts
  left_join(comment_counts, by = "video_id") %>%
  # If a video had no comments in the comment file, replace NA with 0
  mutate(comment_count = replace_na(comment_count, 0)) %>%
  # Remove irrelevant variables
  select(-any_of(c("date", "tags", "comments", "transcript", "transcript_str"))) %>%
  # Create the dichotomous variable (filled with NA for now so you can code it)
  mutate(video_debunking = NA_real_)

# View the prepped data
head(video_prep)

```

```

## # A tibble: 6 x 10
##   ...1 video_id  title      description  views likes dislikes channel_title
##   <dbl> <chr>      <chr>      <chr>      <dbl> <dbl> <lgl>      <chr>
## 1      0 OVSV99PXbbw Why Trump k~ "Universit~ 3.12e5  4327 NA      <NA>
## 2      1 39voRIZciVI Springfield~ "In a one~  3.21e5  5938 NA      <NA>
## 3      2 4IGd0BrrXoI Trump, Harr~ "Former Pr~ 1.86e6 27856 NA      <NA>
## 4      3 f1RF50QrURI Trump ampli~ "Former Pr~ 4.34e4   546 NA      <NA>
## 5      4 GTmdOGut4Gw Springfield~ "A Springf~ 2.37e5  5363 NA      <NA>
## 6      5 L5Gq1Tc63UE Springfield~ "Springfie~ 1.34e6 35919 NA      <NA>
## # i 2 more variables: comment_count <int>, video_debunking <dbl>

```

Coding the video_debunking variable

Since there were only 23 videos, manual coding was a very feasible option, and therefore chosen. This ensures no errors as a result of automation in the variable which is to be created below. The corresponding video_id's were found by opening dataframe 'video_prep' in the environment.

Title	Debunking / Not debunking
Why Trump keeps pushing debunked claims about Haitian immigrants	Debunking
Springfield mayor responds to JD Vance's claims about Haitian immigrants	Not debunking
Trump, Harris face off at the ABC News Presidential Debate	Not debunking
Trump amplifies false claims about Haitian immigrants eating pets	Debunking
Springfield, Ohio, man reported Haitian migrants snatching geese out of park just 2 weeks ago	Not debunking
Springfield resident describes impact of Haitian migrants on community	Not debunking
Springfield neighbors describe immigrant situation; say they believe pet-snatching story	Not debunking
JD Vance exposed for knowingly lying about 'pet-eating' rumors in Springfield, OH	Debunking
'You think this is the last cup of B.S.?': Steele slams Trump campaign for lying about migrants	Debunking

Ohio residents rage at officials over Haitian migrant surge: 'No vacancy'	Not debunking
Trump spreads more lies about Springfield, Ohio	Debunking
Springfield, Ohio resident: This is a dystopian nightmare	Not debunking
Haitian community in Springfield, Ohio, under threat after debunked comments	Debunking
Reporter Presses JD Vance About 'Debunked False Claims' About Haitians In Springfield, Ohio	Debunking
Criminal charges filed by Haitian nonprofit against Trump and JD Vance	Debunking
'NO DOGS OR CATS' Haitian immigrant pleads with Trump and Springfield, Ohio residents	Debunking
Missing cat at center of Springfield conspiracy theory found hiding in basement	Debunking
Trump falsely claims Haitian migrants are eating pets in Ohio	Debunking
JD Vance claims on Ohio's Haitian immigrants eating pets debunked	Debunking
Dana Bash and JD Vance clash over baseless claims about Haitian immigrants (Part 1/2)	Debunking
JD Vance reacts to Springfield father saying he politicized his son's death (Part 2/2)	Not debunking
JD Vance defends amplifying false claims about immigrants	Debunking
Residents in Springfield, Ohio react to accusations of immigrants abducting and eating pets	Not debunking

```

# 1. Manually list the exact video IDs that are DEBUNKING
# (Replace these placeholders with your actual video IDs)
debunking_videos <- c("xc0IkVmiN8o", "flRF50QrURI", "6W0mzgCdNvY", "eyt1X20TKPs", "Ru34gTyHK1w", "OVS99")

# 2. Apply the manual coding and create the factor
video_final <- video_prep %>%
  mutate(
    # First, create the numeric version for your correlation analysis
    video_debunking_num = ifelse(video_id %in% debunking_videos, 0, 1),

    # Next, create the factor version for your tables and summaries
    video_debunking_factor = factor(video_debunking_num,
                                    levels = c(0, 1),
                                    labels = c("debunking", "not debunking"))
  )

# View a quick summary to make sure your factor applied correctly
summary(video_final$video_debunking_factor)

```

```
##      debunking not debunking
##           14              9
```

Summarising engagement

```
# Use the factor version so the table reads "debunking" / "not debunking"
overview_table <- video_final %>%
  select(title, video_debunking_factor, views, likes, comment_count)

kable(overview_table,
      caption = "Overview of Video Engagement",
      format.args = list(big.mark = ","))
```

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 1: Overview of Video Engagement

title	video_debunking_factor	views	likes	comment_count
Why Trump keeps pushing debunked claim about Haitian immigrants	debunking	312,222	4,327	5,132
Springfield mayor responds to JD Vance's claims about Haitian immigrants	not debunking	320,834	5,938	4,070
Trump, Harris face off at the ABC News Presidential Debate	not debunking	1,858,436	27,856	10,803
Trump amplifies false claims about Haitian immigrants eating pets	debunking	43,414	546	820
Springfield, Ohio, man reported Haitian migrants snatching geese out of park just 2 weeks ago	not debunking	236,924	5,363	1,670
Springfield resident describes impact of Haitian migrants on community	not debunking	1,342,405	35,919	9,085
Springfield neighbors describe immigrant situation; say they believe pet-snatching story	not debunking	261,990	4,575	3,581
JD Vance exposed for knowingly lying about 'pet-eating' rumors in Springfield, OH	debunking	280,072	5,830	1,917
'You think this is the last cup of B.S.?': Steele slams Trump campaign for lying about migrants	debunking	442,417	8,761	2,108
Ohio residents rage at officials over Haitian migrant surge: 'No vacancy'	not debunking	1,233,287	34,778	9,843
Trump spreads more lies about Springfield, Ohio	debunking	297,249	3,370	4,536
Springfield, Ohio resident: This is a dystopian nightmare	not debunking	453,853	10,533	144
Haitian community in Springfield, Ohio, under threat after debunked comments	debunking	36,872	459	916
Reporter Presses JD Vance About 'Debunked False Claims' About Haitians In Springfield, Ohio	debunking	82,768	2,238	926
Criminal charges filed by Haitian nonprofit against Trump and JD Vance	debunking	44,919	1,496	1,290
'NO DOGS OR CATS' Haitian immigrant pleads with Trump and Springfield, Ohio residents	debunking	54,222	809	1,349
Missing cat at center of Springfield conspiracy theory found hiding in basement	debunking	332,473	5,631	2,270
Trump falsely claims Haitian migrants are eating pets in Ohio	debunking	88,241	986	1,974

title	video_debunking_factor	views	likes	comment_count
JD Vance claims on Ohio's Haitian immigrants eating pets debunked	debunking	86,482	608	1,368
Dana Bash and JD Vance clash over baseless claims about Haitian immigrants (Part 1/2)	debunking	838,589	12,164	1,536
JD Vance reacts to Springfield father saying he politicized his son's death (Part 2/2)	not debunking	379,162	6,603	101
JD Vance defends amplifying false claims about immigrants	debunking	195,846	3,467	3,969
Residents in Springfield, Ohio react to accusations of immigrants abducting and eating pets	not debunking	243,855	1,296	1,617

Analysing the correlation

```
# Use the numeric version so the correlation math works
correlation_data <- video_final %>%
  select(video_debunking_num, views, likes, comment_count) %>%
  drop_na()

correlation_results <- correlation_data %>%
  correlate(method = "spearman", diagonal = 1)
```

```
## Correlation computed with
## * Method: 'spearman'
## * Missing treated using: 'pairwise.complete.obs'
```

```
kable(correlation_results,
      digits = 3,
      caption = "Spearman Correlation Matrix")
```

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 2: Spearman Correlation Matrix

term	video_debunking_num	views	likes	comment_count
video_debunking_num	1.000	0.524	0.551	0.215
views	0.524	1.000	0.941	0.534
likes	0.551	0.941	1.000	0.462
comment_count	0.215	0.534	0.462	1.000